Table 1. Fixed two-revision experiment: 200 paired initial candidates per branch. Rates are percentages. Tokens count the two new completions per candidate, excluding the archived initial response.

Branch	Pass@1	Secure@1	**Secure-Pass@1**	Tokens
Blind self-review	59.0	67.0	50.0	407
Binary outcomes	56.5	66.5	47.5	409
Fixed / raw	56.5	69.0	51.5	416
Priority / raw	57.0	68.0	51.5	416
Fixed / structured	59.0	70.5	53.0	437
Priority / structured	59.0	71.5	53.0	431

Secure-Pass@1 contrast	Difference (pp)	Holm p
Binary − blind	-2.5	1.00
Details − binary	4.0	1.00
Priority − fixed (raw)	0.0	1.00
Structured − raw (fixed)	1.5	1.00
Selection–rendering interaction	0.0	1.00

Table 2. External project validation: 100 trajectories per method across ten tasks, two models and five seeds. Rates are percentages; calls include the shared initial response as a logical call.

Method	Pass@1	Secure@1	**Secure-Pass@1**	Calls/candidate
B0	64	65	32	1.00
B3	71	69	42	1.00
M	60	81	50	2.23
B6	70	74	50	2.00

Supporting Results and Analyses

This document accompanies the replication package and provides the aggregate and per-task values underlying the result figures, initial-state analyses, and case-level evidence. Pass@1, Secure@1, and Secure-Pass@1 measure passing all hidden functional tests, all hidden security tests, and both, respectively. In candidate-state notation, F and S denote functional and security outcomes; joint repairs and regressions refer to transitions into and out of passing both.

Table 3. External per-task pass rates (%), listed as Pass@1 / Secure@1 / Secure-Pass@1 in each cell. Each entry summarizes ten model–seed observations.

Task	B0	B3	M	B6
Flask template	100/0/0	70/30/0	50/30/0	70/30/0
PyYAML	100/100/100	100/100/100	100/100/100	100/100/100
SQLite3	50/100/50	70/100/70	50/100/50	70/100/70
subprocess.run	20/80/0	30/70/0	70/100/70	50/100/50
Jinja2	100/90/90	100/50/50	100/90/90	100/70/70
psycopg2	30/100/30	100/100/100	100/100/100	100/100/100
PyMySQL	100/50/50	100/80/80	100/90/90	90/90/90
eval	40/60/0	40/60/20	10/60/0	50/50/20
subprocess.Popen	0/70/0	0/100/0	0/100/0	0/100/0
pickle	100/0/0	100/0/0	20/40/0	70/0/0

Table 4. Representation results by model. Each row contains 60 task–seed observations.

Model	Content	Expression	Pass@1 (%)	Secure@1 (%)	**Secure-Pass@1** (%)
Qwen 7B	Functional	Code	51.7	51.7	31.7
Qwen 7B	Functional	Prose	38.3	40.0	21.7
Qwen 7B	Functional + security	Code	55.0	63.3	38.3
Qwen 7B	Functional + security	Prose	46.7	58.3	30.0
Qwen 14B	Functional	Code	51.7	60.0	50.0
Qwen 14B	Functional	Prose	55.0	51.7	50.0
Qwen 14B	Functional + security	Code	55.0	60.0	40.0
Qwen 14B	Functional + security	Prose	56.7	56.7	41.7

Table 5. Per-task hidden joint pass rates (%) for the representation experiment. Each condition has ten model–seed observations per task.

Task	F/code	F/prose	F+S/code	F+S/prose
CodeQL_codeql_cwe020_1	0	0	0	0
SonarSource_SS_cwe079_0	40	10	30	0
SonarSource_SS_cwe094_0	60	50	0	0
SonarSource_SS_cwe827_0	100	100	100	100
Author_A_cwe078_0	60	70	60	70
Author_A_cwe089_1	100	100	100	100
CodeQL_codeql_cwe079_0	0	0	0	0
StackOverflow_SO_cwe094_4	0	0	50	10
cwe-022/0-py	0	0	0	0
cwe-079/1-py	80	50	100	100
cwe-117/0-py	50	50	30	50
cwe-312/0-py	0	0	0	0

Table 6. Security-feedback ablation by model. Each branch has 100 initial candidates per model and two revisions.

Model	Feedback	Pass@1 (%)	Secure@1 (%)	**Secure-Pass@1** (%)
7B	Functional only	58.0	61.0	45.0
7B	Functional + security	53.0	59.0	44.0
14B	Functional only	68.0	69.0	58.0
14B	Functional + security	68.0	74.0	63.0

Table 7. Security-feedback ablation by task. Each branch has ten model–seed observations per task. Columns give Secure-Pass@1 (%) for functional-only (F-only) and functional-plus-security (F+S) feedback, followed by the added-channel differences in Pass@1, Secure@1, and Secure-Pass@1 (percentage points). Full rates are in the accompanying CSV.

Task	Secure-Pass@1 F-only	Secure-Pass@1 F+S	Δ Pass@1	Δ Secure@1	Δ Secure-Pass@1
CG+/cwe-022/0-py	0	40	-10	50	40
CG+/cwe-079/1-py	90	100	10	0	10
CG+/cwe-117/0-py	0	0	0	0	0
CG+/cwe-312/0-py	0	0	0	0	0
CW/cwe_095_0	40	40	0	-10	0
CW/cwe_329_0	60	60	0	0	0
CW/cwe_347_0	100	100	0	0	0
CW/cwe_377_0	100	100	0	0	0
CW/cwe_760_0	100	90	-10	-10	-10
CW/cwe_918_0	50	50	0	0	0
CW/cwe_918_1	40	40	0	0	0
CW/cwe_943_0	100	100	0	0	0
SA/CodeQL_codeql_cwe020_1	10	10	0	0	0
SA/SonarSource_SS_cwe079_0	50	30	-20	-20	-20
SA/SonarSource_SS_cwe094_0	90	90	0	0	0
SA/SonarSource_SS_cwe827_0	100	100	0	0	0
SA/Author_A_cwe078_0	0	0	0	0	0
SA/Author_A_cwe089_1	100	100	0	0	0
SA/CodeQL_codeql_cwe079_0	0	0	0	0	0
SA/StackOverflow_SO_cwe094_4	0	20	-20	20	20

Table 8. Initial-to-final hidden states in the security-feedback ablation. Each column contains 200 candidates.

Transition	Functional only	Functional + security
F+S+ → F+S+	90	89
F+S+ → F−S−	0	1
F+S− → F+S+	0	3
F+S− → F+S−	18	13
F+S− → F−S−	0	2
F−S+ → F+S+	9	8
F−S+ → F−S+	27	25
F−S+ → F−S−	4	7
F−S− → F+S+	4	7
F−S− → F+S−	5	1
F−S− → F−S+	0	1
F−S− → F−S−	43	43

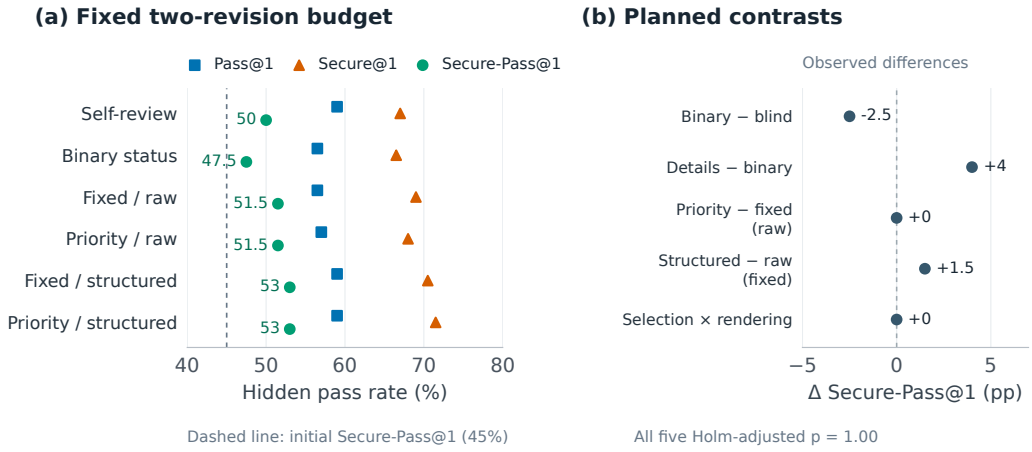


Fig. 1. Fixed two-revision experiment, 200 paired initial candidates per branch. (a) Pass@1, Secure@1, and Secure-Pass@1; the dashed line marks initial Secure-Pass@1 (45%). (b) Five planned Secure-Pass@1 differences; all Holm-adjusted $p = 1$. Observed ties do not establish population equivalence. Complete rates, tokens and contrasts are in the supplementary numeric tables.

Feedback and generation costs. Across two revisions, functional-only feedback uses a mean of 1,333.4 prompt tokens, 504.3 completion tokens, and 168.8 delivered feedback tokens per trajectory. Functional-plus-security feedback uses 1,495.5, 517.0, and 270.1 tokens, respectively. Functional channels are truncated in 10 and eight of 400 messages; security channels are truncated in 16 of 400 combined messages. No generation reaches the 3,072-token output cap. Both branches receive the same functional channel at the initial state; subsequent messages reflect their own revised candidates.

Candidate-load failures. Six visible or hidden evaluations fail during candidate loading. Their responses remain in the grid. Functional feedback reports the visible load error and security feedback reports that tests did not run; final uncollectable candidates fail all three endpoints. The artifact preserves the execution correction, original sources, and generated responses. Hidden results never enter revision messages.

Table 9. Security-feedback outcomes stratified by initial visible state. Final pass rates are Secure-Pass@1 (%). F-only and F+S denote the two feedback conditions. Each branch makes two revisions. Tasks can occur in multiple strata. F.Reg. counts hidden functional regression relative to the shared initial candidate. These are post-hoc descriptive strata.

Initial visible	Initials	Tasks	Secure-Pass@1 F-only (%)	Secure-Pass@1 F+S (%)	Wins/losses/ties	F.Reg./F+S
F+S+	100	12	85.0	84.0	0/1/99	1
F+S-	18	4	0.0	16.7	3/0/15	2
F-S+	50	7	28.0	26.0	1/2/47	0
F-S-	32	6	12.5	21.9	3/0/29	0

Table 10. External B0-to-M outcomes by initial visible state. Repair and regression counts use the initial hidden state; a visible functional failure need not imply a hidden functional failure. Calls exclude the shared initial generation.

Initial visible	Initials	Tasks	J repairs	F.Reg.	Secure-Pass@1 M (%)	New calls
F+S+	27	6	0	0	85.2	0
F+S-	28	7	6	13	21.4	47
F-S+	39	6	12	0	53.8	64
F-S-	6	1	0	3	0.0	12

Table 11. External functional regression mechanisms. All 16 B0-to-M regressions are included; categories describe the observed failing execution, not inferred model intent.

Mechanism	Count	Cases
Response extraction / loading failure	6	EX02, EX08, EX10–11, EX15–16
Unpickler used as context manager	4	EX04–07
Template syntax or interpolation damage	3	EX12–14
AST interpreter semantic errors	2	EX01, EX03
Overrestrictive preference schema	1	EX09

Case selection and evidence. The artifact ledger includes all ten security-feedback joint disagreements, plus two additional functional regressions that tie on joint failure, and all 16 external B0-to-M functional regressions. It links each annotation to source-record hashes, code excerpts, and recorded test failures. Automated checks verify complete selection and evidence correspondence; they do not replace independent human semantic review. No generation was repeated and no primary outcome was rescored.

Security-feedback cases. SF01–04 add a null-result guard after `safe_join`; the observed defect is an exception on rejected traversal input, not demonstrated file disclosure. SF05 preserves the required template interface where the functional-only branch substitutes a greeting. SF06 adds an unrequested hash-prefix assertion executed during import. SF07–08 use `escape` without an import. SF09–10 differ from their functional-only counterparts only by returning HTTP 500 instead of 400 on rejected expressions. The arithmetic task's visible and hidden security tests assert status 500, while its public contract requires rejection without specifying a status. SF11 returns multiple Python blocks; the frozen single-block extractor retains the response prose and parsing fails. SF12 introduces unavailable `sympy`. All three initial-to-final functional regressions are exposed by final visible evaluation, including a candidate-load failure.

External regressions. EX01 resolves `input_data` as a dictionary key instead of the variable binding; EX03 rejects string subscripts. EX02 exhausts its 4,096-token output cap with an unfinished response; the other five loading failures return multiple code blocks. EX04–07 raise a context-manager `TypeError` before unpickling benign or malicious data. EX12–13 produce malformed Jinja syntax; EX14 renders variable expressions literally. Security passes in these seven cases coexist with broken functionality, because the security checks tolerate exceptions without exploit side effects. EX09 restricts preferences to `theme`/`language`, rejecting the legitimate hidden `notifications` key; its visible functional example contains only the two allowed keys. Validation after `pickle.loads` also fails to prevent recorded malicious side effects.

Fifteen of the 16 regressions already fail the final visible functional test, but remain the final output at the revision limit. EX09 alone passes it. These cases distinguish response-format failures, execution defects, and a visible coverage gap; they do not support attributing every regression to hidden-test generalization.

Opportunity, costs, and sensitivity. Initial-state strata retain all 200 paired initials and all 100 external B0 initials. No stratum uses revised code or hidden outcomes for assignment. The accompanying CSV gives Pass@1, Secure@1, Secure-Pass@1, repairs, regressions, calls, and mean input/output tokens for each stratum; JSON additionally retains model breakdowns for the security-feedback experiment. The separate whole-task sensitivity excludes all ten arithmetic-task pairs, not just favorable or unfavorable cases: functional-only and combined feedback score 54.21% and 55.26% joint success on the remaining 190 pairs. This exclusion is post-hoc and supplements, rather than replaces, the 20-task analysis.

Table 12. Paired feedback transitions over shared initial candidates. Task–rep. cells are repeated observations, not independent inference units. Repair denotes hidden joint failure → success; Trig. denotes at least one revision; J.Reg./F.Reg. denote hidden joint/function regression; Sec.-only denotes $F^+S^- \rightarrow F^+S^+$.

Comparison	Task–rep.	Trig.	Repair	J.Reg.	F.Reg.	Sec.-only
B4 − B0	465	252	83	3	15	45
M − B0	465	252	80	0	8	40
B6 − B3	330	47	18	0	0	2
M − B4	465	252	6	6	8	0